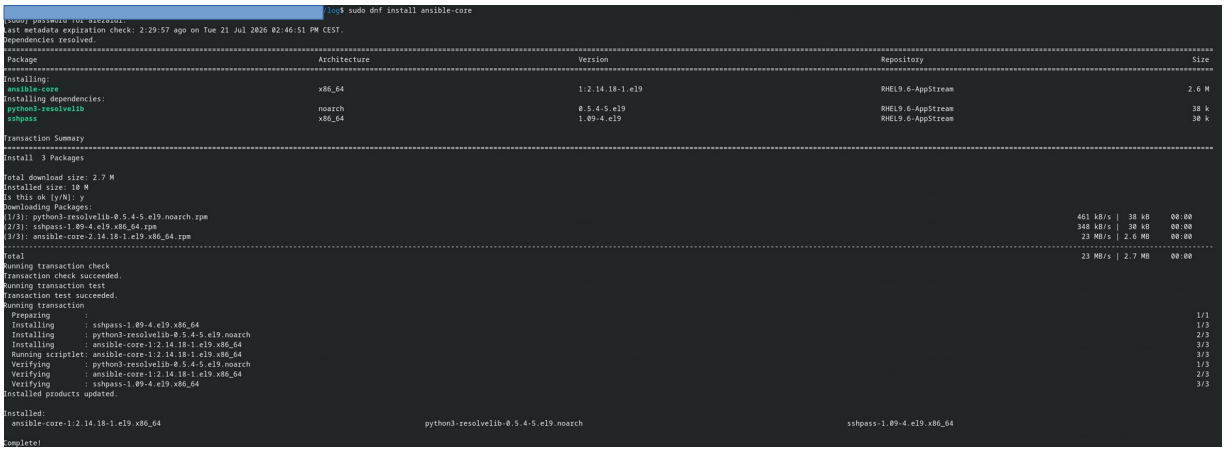
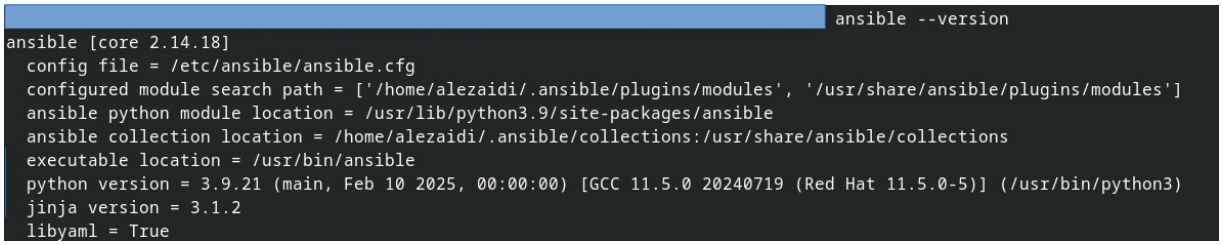
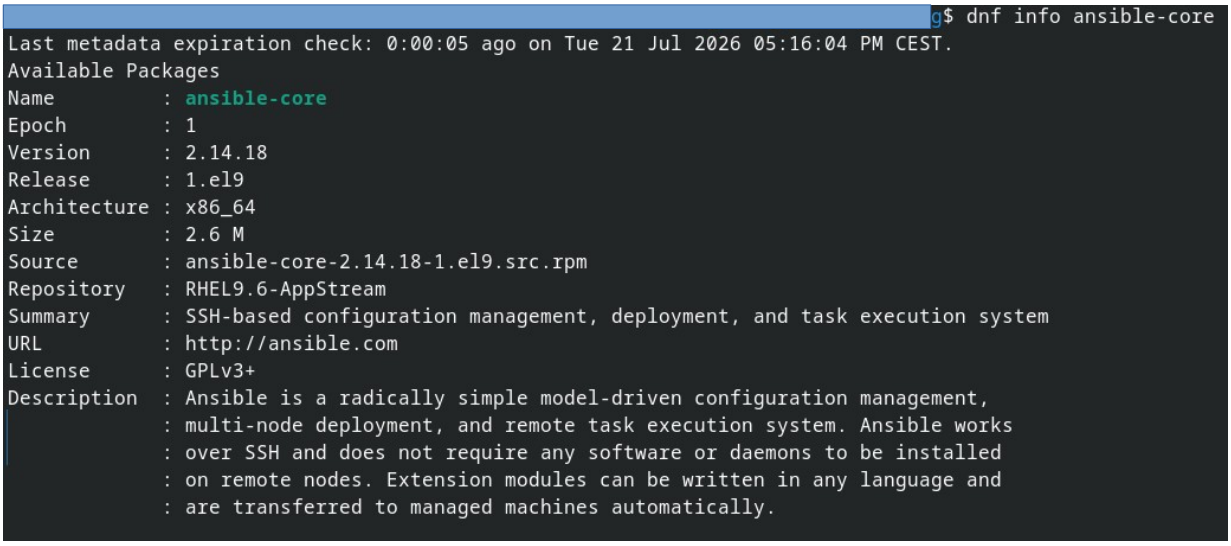
Ansible


log$ sudo dnf install ansible-core
Last metadata expiration check: 2:29:57 ago on Tue 21 Jul 2026 02:46:51 PM CEST.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
ansible-core x86_64 1:2.14.18-1.el9 RHEL9.6-AppStream 2.6 M
Installing dependencies:
python3-resolvelib noarch 0.5.4-5.el9 RHEL9.6-AppStream 38 k
sshpass x86_64 1.09-4.el9 RHEL9.6-AppStream 38 k
Transaction Summary
Install 3 Packages
Total download size: 2.7 M
Installed size: 10 M
Is this ok [y/N]: y
Downloading Packages:
(1/3): python3-resolvelib-0.5.4-5.el9.noarch.rpm 461 kB/s | 38 kB 00:00
(2/3): sshpass-1.09-4.el9.x86_64.rpm 348 kB/s | 38 kB 00:00
(3/3): ansible-core-2.14.18-1.el9.x86_64.rpm 23 MB/s | 2.6 MB 00:00
Total 23 MB/s | 2.7 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : sshpass-1.09-4.el9.x86_64 1/3
Installing : python3-resolvelib-0.5.4-5.el9.noarch 2/3
Installing : ansible-core-1:2.14.18-1.el9.x86_64 3/3
Running scriptlet: ansible-core-1:2.14.18-1.el9.x86_64 3/3
Verifying : python3-resolvelib-0.5.4-5.el9.noarch 1/3
Verifying : ansible-core-1:2.14.18-1.el9.x86_64 2/3
Verifying : sshpass-1.09-4.el9.x86_64 3/3
Installed products updated.
Installed:
ansible-core-1:2.14.18-1.el9.x86_64 python3-resolvelib-0.5.4-5.el9.noarch sshpass-1.09-4.el9.x86_64
Complete!


ansible --version
ansible [core 2.14.18]
config file = /etc/ansible/ansible.cfg
configured module search path = ['/home/alezaidi/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
ansible python module location = /usr/lib/python3.9/site-packages/ansible
ansible collection location = /home/alezaidi/.ansible/collections:/usr/share/ansible/collections
executable location = /usr/bin/ansible
python version = 3.9.21 (main, Feb 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3)
jinja version = 3.1.2
libyaml = True


g$ dnf info ansible-core
Last metadata expiration check: 0:00:05 ago on Tue 21 Jul 2026 05:16:04 PM CEST.
Available Packages
Name : ansible-core
Epoch : 1
Version : 2.14.18
Release : 1.el9
Architecture : x86_64
Size : 2.6 M
Source : ansible-core-2.14.18-1.el9.src.rpm
Repository : RHEL9.6-AppStream
Summary : SSH-based configuration management, deployment, and task execution system
URL : http://ansible.com
License : GPLv3+
Description : Ansible is a radically simple model-driven configuration management,
: multi-node deployment, and remote task execution system. Ansible works
: over SSH and does not require any software or daemons to be installed
: on remote nodes. Extension modules can be written in any language and
: are transferred to managed machines automatically.

Ansible Project Structure:

folders are in red color. Files are in green color

- ansible-project-1
  - ansible.cfg
  - inventory
    - hosts
  - roles
    - installRPM
      - tasks
        - main.yml
    - webserver
      - tasks
        - main.yml
    - create-directories
      - tasks
        - main.yml
  - playbook.yml

```
[local]
Localhost ansible_connection=local

[lab-computers]
Lab-vm-1 ansible_host=192.168.1.0 ansible_user=aley
Lab-vm-2 ansible_host=192.168.2.0 ansible_user=aley
```

```
- name: Install basic packages
  ansible.builtin.dnf:
    name:
      - rsync
      - vim-enhanced
      - kdiff3
      - tmux
    state: installed
```

```
- name: Install webserver apache2
  ansible.builtin.dnf:
    name: apache2
    become=yes
    uppdate_cache=yes
    state=latest
```

```
- name: practice directories
  hosts: lab-computers

  tasks:
    - name: Create Project Directory
      ansible.builtin.file:
        path: /tmp/ansible_lab
        state: directory
        mode: '0755'

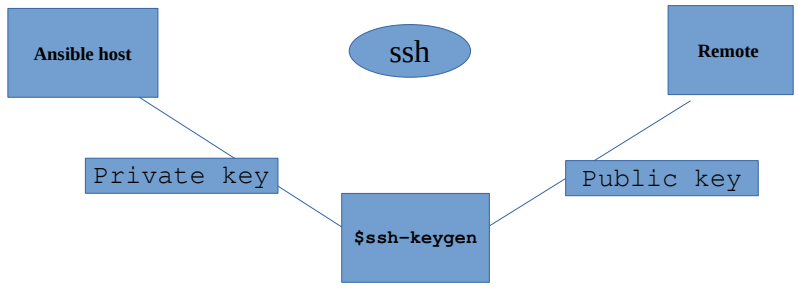
    - name: Create log Directory
      ansible.builtin.file:
        path: /tmp/ansible_lab/logs
        state: directory

    - name: Create File
      ansible.builtin.file:
        path: /tmp/ansible_lab/hurra.txt
        state: touch
```

```
- name: Perform ansible operation
  hosts: all
  remote_user: aley
  roles:
    - role: installRPM
    - role: webserver
    - role: create-directories
```

To execute Ansible Playbook:
ansible-playbook –inventory inventory/hosts playbook.yml

using ssh keys to connect from ansible host machine to remote machine



```
                                ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/araza/.ssh/id_rsa): /home/araza/.ssh/ansible-test
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/araza/.ssh/ansible-test
Your public key has been saved in /home/araza/.ssh/ansible-test.pub
The key fingerprint is:
SHA256:7ivFaYXGsWhZGoiPaA1JCH/JdNzm9SMXIiV+B6T4+gI araza@boeow728
The key's randomart image is:
+---[RSA 3072]----+
|=....o...o+      |
|.+.o.ooo*oo..    |
| .+o+ .Xo*.o..   |
|...o. =.*.o.+    |
|.    . oSo o .   |
|    E  o=        |
|     ..o.        |
|      oo         |
|       o+.       |
+----[SHA256]-----+
```

copy public key to the remote machine

```
                          $ ssh-copy-id -i /home/araza/.ssh/ansible-test.pub alezaidi@192.168.10.1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/araza/.ssh/ansible-test.pub"
```

now in your host file, on the line where you mentioned remote machine, replaced with following:

~~Lab-vm-1 ansible_host=192.168.1.0 ansible_user=aley~~

```
Lab-vm-1 ansible_host=192.168.1.0 ansible_ssh_private_key_file=/home/araza/.ssh/ansible-test
```

now when you execute ansible playbook, it will not ask for password

some most frequently used ansible and their parameters

```
 ansible.builtin.file:
    path: /tmp/ansible_lab
    state: directory
    mode: '0755'


ansible.builtin.ping:

ansible.builtin.template
- name www index.html file
  template. src=updates.html destination /usr/share/nginx/html/index.html
  notify:
    - Restart nginx
```

notify command is to call a handler which must be defined in main play book.yml file

playbook.yml

```
- name: Perform ansible operation
  hosts: all
  remote_user: aley
  roles:
    - role: installRPM
    - role: webserver
    - role: create-directories
  handlers:
    - name: Restart nginx
      service:
        name: ngix
        state: restarted
```